EXPERIMENT – 9

AIM :

To create a Railway Booking table containing at least 5,000 records and benchmark query performance before and after creating single and composite indexes. To compare query execution plans using the EXPLAIN command.

SOURCE CODE :

```
-- Create Railway Booking Table

CREATE TABLE RailwayBooking (
   BookingID INT PRIMARY KEY,
   PassengerName VARCHAR(50),
   TrainID INT,
   JourneyDate DATE,
   SeatNo INT,
   Status VARCHAR(20)
);

-- Insert 5000 Records

DELIMITER //

CREATE PROCEDURE InsertBookings()
BEGIN
   DECLARE i INT DEFAULT 1;

   WHILE i <= 5000 DO
      INSERT INTO RailwayBooking
      VALUES (
         i,
         CONCAT('Passenger', i),
         100 + (i % 20),
         DATE_ADD('2026-01-01',
            INTERVAL (i % 365) DAY),
         i % 100 + 1,
         IF(i % 2 = 0, 'Confirmed', 'Waiting')
      );

      SET i = i + 1;
   END WHILE;
END //

DELIMITER ;

CALL InsertBookings();
```

```
-- Verify Total Records

SELECT COUNT(*) FROM RailwayBooking;

-- 1. Query Before Indexing

EXPLAIN
SELECT * FROM RailwayBooking
WHERE TrainID = 105;

SELECT * FROM RailwayBooking
WHERE TrainID = 105;

-- 2. Create Single Index

CREATE INDEX idx_train
ON RailwayBooking(TrainID);

-- Query After Single Index

EXPLAIN
SELECT * FROM RailwayBooking
WHERE TrainID = 105;

SELECT * FROM RailwayBooking
WHERE TrainID = 105;

-- 3. Create Composite Index

CREATE INDEX idx_train_date
ON RailwayBooking(TrainID, JourneyDate);

-- Query After Composite Index

EXPLAIN
SELECT * FROM RailwayBooking
WHERE TrainID = 105
AND JourneyDate = '2026-01-06';

SELECT * FROM RailwayBooking
WHERE TrainID = 105
AND JourneyDate = '2026-01-06';

-- 4. Display Indexes

SHOW INDEX FROM RailwayBooking;
```

-- 5. Check Query Execution Plan

```
EXPLAIN
SELECT * FROM RailwayBooking
WHERE TrainID = 105
AND JourneyDate = '2026-01-06';
```

OUTPUT :

